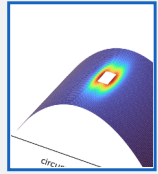


STRUCTURE-SPECIFIC

SHARED DECISION CORE (structure-agnostic)



DSPSS field

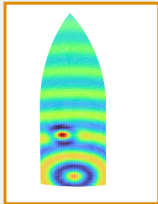
Stage 1 detection

Interstage

perforated CFRP

surface-stress GNN

DSPSS · F1 0.79



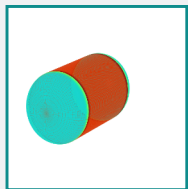
guided-wave field

Fairing

honeycomb skin-core

guided-wave GNN

Lamb · F1 0.86



hoop-stress field (σ_θ)

SRB motor case

filament-wound CFRP

hoop-stress screen

σ_θ · burst / debond

merge

Stage 0/2

anomaly screen + characterisation

AUROC 0.85

Stage 3 prognosis

pluggable: phase-field / debond / burst

Clearance

OK

REPAIR

RETIRE

85% / 92.5% acc
0% dangerous-miss

Stage 4 fleet learning

Stage 5 design feedback

ECE 0.13→0.03 · sim-to-real FPR 14→5%

design-feedback loop